NB-IoT RACH Revision — Problem Definition, Literature Review & Research Gaps

1. Reviewer comments and recommended changes

The reviewers see the smart-meter communication problem as useful. Their main concerns are that the paper's contribution is not clearly different from earlier work, the comparisons are inconsistent, and important explanations are missing.

The comments below come from the supplied DELCON review PDF. The recommendations explain how to address them in the paper. **Code review and code changes are outside this document.**

A. The comparisons do not match

Reviewer's comment: The experimental setup, results table, and figures show different methods. For example, the setup mentions single-parameter and dual-parameter RL, while the figures show DQN, DDQN, and PPO.

What this means: Readers cannot tell which methods were actually compared.

Recommended changes:

- Choose one clearly defined set of comparison methods.
- Use the same names in the experimental setup, tables, figures, and discussion.
- Explain what each method controls and how it was evaluated.
- Include single-control and two-control comparisons if the paper claims that controlling three parameters together is better.

B. Proxy results do not prove that the proposed method is better than deep RL

Reviewer's comment: Figure 4 calls the deep-RL results "DRL proxies."

What this means: An approximate or illustrative result is not the same as testing the actual algorithm.

Recommended changes:

- Compare against actual evaluated methods, not estimated behaviour presented as algorithm results.
- If actual DQN, DDQN, or PPO results are unavailable, remove claims of outperforming those methods.
- A smaller, honest comparison is stronger than a larger comparison that cannot be supported.

C. Some performance measures are unclear

Reviewer's comment: Collision probability and success probability are shown as complements, and the quantity `e` is not defined.

What this means: Some columns repeat the same information, while another column has no clear meaning.

Recommended changes:

- Define every measure before presenting results.
- Explain whether "success" means a successful attempt, a completed access procedure, or delivery of a meter message. These are different outcomes.
- If success and collision are the only possible outcomes for the same set of attempts, explain that their probabilities add to one. Do not present them as two independent improvements.
- Define `e`, including its unit. If it is only an estimate of transmission effort, do not call it measured energy consumption.
- Along with average delay, report how many requests finish within the target time and how many remain unfinished.

D. The new contribution is not strong or clear enough

Reviewer's comment: The work appears to be a small variation of earlier Q-learning approaches for random access.

What this means: Using Q-learning, or adding another control parameter, is not enough by itself to establish a strong contribution.

Recommended changes:

- Explain what earlier methods already do before stating what this paper adds.
- Avoid broad statements such as “existing methods optimise only one parameter.” Relevant earlier studies already consider several controls.
- Focus on a clear question: **Can a small, network-side controller help more smart-meter requests finish within a target time, using a fair amount of network resources?**
- Show the separate benefit of joint control, realistic smart-meter traffic, and low controller cost. Treat these as contributions to demonstrate, not benefits already proved.

E. The reward calculation is not explained

Reviewer’s comment: Figure 2 includes a “compute reward” step, but the text does not explain it properly. The link from reward calculation to the learning update is also missing.

What this means: Readers cannot understand exactly what behaviour the learning agent is encouraged to follow.

Recommended changes:

- Give the exact reward equation and explain each term in plain language.
- State what gives a higher reward and what gives a penalty.
- Explain the order: observe the network → choose settings → observe the outcome → calculate reward → update learning.
- Redraw the workflow so every step is connected correctly.

F. Algorithm 1 is missing

Reviewer’s comment: The text refers to Algorithm 1, but the algorithm is not included.

Recommended changes: Include a short, complete step-by-step description of the learning procedure. Make sure it agrees with the equations and workflow

figure.

G. Presentation errors make the paper difficult to trust

Reviewer's comments: There is a broken figure reference, an unfinished `[Rxx]` citation, spacing mistakes, and a cropped Figure 4. Reviewer B also expresses concern about careless AI-assisted presentation.

Recommended changes:

- Remove all broken references and unfinished placeholders.
- Check every citation against the paper it is supposed to support.
- Redraw unclear diagrams with readable labels and correct arrows.
- Check that every figure fits inside the final PDF and remains readable at normal size.
- Read the final exported PDF carefully, not just the editable document. Every claim and figure should be checked by the authors.

Citation caution: Reviewer A suggests using reference [13] for LE-URC, but [13] in the supplied manuscript is a different Q-learning paper. Verify the correct LE-URC source rather than simply changing the reference number. Jiang et al.'s NB-IoT paper discusses LE-URC.[1]

H. The indicator function needs a clearer explanation

Reviewer's comment: Reviewer B questions the indicator-function notation in the SLA equation.

Clarification: An indicator function is valid mathematical notation. It means **1 when a condition is true and 0 when it is false**. Its use is not automatically an error.

Recommended changes: Explain it in those simple words. Also state exactly what is being counted: delayed requests, failed requests, or time intervals. If the aim is to measure missed message deadlines, the measure should count requests that miss those deadlines.

2. Updated problem statement, literature review, and research gaps

2.1 The problem in simple words

Many smart meters share the same NB-IoT network. They usually send readings at regular times. An event, such as power restoration or a utility request, can cause many meters to try to connect at nearly the same time.

The network has limited opportunities for these connection attempts. When too many meters compete, their attempts can collide. They must wait and try again, which increases delay.

Three settings influence this process:

Setting	Simple meaning	Main trade-off
Access Class Barring (ACB)	Decides which devices may try to access the network now	Allowing too many can increase collisions; allowing too few can delay useful messages.
Backoff setting	Decides how long a device waits before retrying after an unsuccessful attempt	A short wait can cause repeated congestion; a long wait can delay completion.
NPRACH resource configuration	Decides how many radio resources are available for initial access	More access resources can help connection attempts, but leave fewer resources for other transmissions.

These settings affect one another. The research should investigate how to coordinate them, rather than assume that improving one measure—such as collision rate—automatically improves service to meters.

2.2 Updated problem statement — ready to adapt for the paper

> Dense NB-IoT smart-meter networks must support both regular meter reports and sudden increases in access requests during events such as power restoration and demand-response communication. When many meters attempt access together, the limited random-access resources can become congested, causing repeated attempts and longer waiting times. Earlier studies already use adaptive barring, backoff control, resource allocation, and reinforcement

> learning to address this problem. This study therefore investigates a more specific question: can a lightweight network-side controller coordinate access admission, retry timing, and NPRACH resources so that more smart-meter requests complete within a defined target time? The study will consider both completed and unfinished requests, use clearly described smart-meter traffic, and compare methods under the same resource limits. Its aim is to establish the benefit and cost of this coordinated control, rather than claim that Q-learning or joint optimisation is new by itself.

Important boundaries:

- Specify whether the study measures initial preamble success, completion of network access, or actual message delivery. Do not use these terms interchangeably.
- Treat the 30-second threshold as a study assumption unless an applicable utility requirement supports it.
- Do not promise that every request will meet its deadline when demand is greater than the network can serve.

2.3 Updated literature review

A. Adaptive access control is already well studied

Duan et al.'s D-ACB study adjusts access barring according to congestion. It also considers fixed and changing preamble allocations. This shows that useful congestion control does not always require machine learning, and that earlier methods are not all static.[2]

Meaning for this paper: Compare with a well-chosen adaptive method, not only a weak fixed configuration.

B. Q-learning and multi-parameter NB-IoT control are not new

Tello-Oquendo et al. use reinforcement learning to adjust access barring in LTE-A. Jiang et al. study tabular Q-learning, simpler approximations, and deep Q-learning for NB-IoT. Jiang's work also examines several configuration parameters across multiple coverage groups.[3][1]

Meaning for this paper: Do not claim that using Q-learning or controlling several NB-IoT settings is the main novelty. Explain what the proposed combination achieves for the chosen smart-meter problem.

C. Earlier work also considers backoff and combined control

Jiang et al.'s decoupled-learning study considers access barring, backoff, and distributed queuing, together with success, delay, and energy objectives. Chen's thesis also describes deep-RL control of access barring and backoff for sudden device activation and different service requirements. Therefore, the claim that backoff has been ignored by all earlier studies is not supported by the supplied literature.[4][5]

Bui and Pham jointly adjust the barring factor and mean barring time to balance delay and energy. However, **barring time and retry backoff are different settings** and should not be treated as the same parameter. Hadjadj-Aoul and Ait-Chellouche use deep RL to control blocking in NB-IoT under uncertain demand.[6] [7]

Meaning for this paper: State exactly which settings earlier methods control and which settings this study controls. "Joint optimisation" is too broad unless the actual differences are explained.

D. Traffic prediction can help, but prediction alone does not prove better communication

Mehri et al. predict traffic and congested periods using an LSTM/DenseNet-based approach. Jiang et al. also study linking traffic prediction with access-control decisions. Thus, prediction and prediction-assisted control already have relevant prior work.[8][9]

Meaning for this paper: If prediction is included, show that it helps requests finish sooner or more reliably. Accurate predictions alone are not enough.

E. Smart-meter-specific studies already exist

Abbas et al. study NB-IoT smart-meter networks and use regular reporting patterns to improve scheduling. Stusek et al. examine NB-IoT performance and

capacity for smart-metering use cases, including early data transmission. Therefore, the earlier review's statement that no study considers smart-meter traffic should be removed.[10][11]

Meaning for this paper: Explain the particular traffic combination being studied—regular reports plus event-driven requests—and test different levels of synchronisation and demand. The gap is not simply "smart meters have not been studied."

F. Real network conditions and practical costs matter

Martiradonna et al. study and validate models of the NB-IoT random-access procedure. Caso et al. use real network measurements and show that access performance depends on coverage, deployment, and operator settings. These studies show why a simple collision model should not automatically be treated as proof of real-world deployment performance.[12][13]

Alcaraz et al.'s 2025 model-based RL paper is another important comparator. Its publisher abstract describes joint adjustment of NPRACH settings and coverage-level thresholds, with an emphasis on efficient learning. Only the abstract and introduction were available in the retrieved source, so detailed comparisons require its full text.[14]

Meaning for this paper: Measure the proposed controller's memory and decision time. Do not assume that every deep-RL method needs a GPU or that a small table alone proves deployment readiness.

2.4 Updated research gaps

These are **specific questions this revised study can investigate**, not claims that no researcher anywhere has considered them.

Research gap	Simple explanation	What this study should demonstrate
1. Useful service, not only successful attempts	A method can have fewer collisions while leaving many meters waiting.	Whether more generated requests finish within the target time, including a clear count of unfinished or late requests.

Research gap	Simple explanation	What this study should demonstrate
2. Benefit of the exact three-control combination	Earlier work already controls multiple settings, but that does not prove the value of this particular combination for the chosen workload.	Whether admission + retry timing + NPRACH control performs better than properly tuned one-control and two-control alternatives under the same resource limit.
3. Performance across different smart-meter events	A method that works for one burst may not work for another.	Whether it remains useful when reporting times, event sizes, burst duration, and the number of affected meters change.
4. Small controller with a worthwhile trade-off	Using less memory is useful only if communication performance remains acceptable.	The measured trade-off between on-time completion, memory use, decision time, and learning effort.
5. Practical network information and update timing	The base station may not know the exact number of waiting devices, and new settings may take time to reach meters.	Whether the controller still helps with imperfect information and realistic delays between configuration changes.

2.5 Research-gap paragraph — ready to adapt for the paper

> Earlier studies already address adaptive access barring, backoff, joint resource control, traffic prediction, and smart-meter scheduling. However, these ideas do not by themselves establish the performance of a small joint controller for the specific smart-meter workload considered here. This study will examine whether coordinated admission, retry timing, and NPRACH configuration can help more requests finish within a target time under the same network resource limits. It will also examine how the benefit changes with different traffic events, imperfect network information, and controller cost. Comparisons with clearly defined existing methods and simpler versions of the proposed controller will be used to establish which parts provide a real benefit.

2.6 Main message to carry into the revised paper

> **The contribution should not be “we use Q-learning to reduce collisions.” It should be “we test whether a small, coordinated controller can provide more**

> **timely service to smart meters, and clearly show when it works, what it costs, and where its limits are.”**

Evidence limit: This review is based on the two supplied PDFs and a targeted review of relevant repository literature, with the Alcaraz publisher summary as a supplement. It is not an exhaustive search proving a “first-ever” contribution. No new experiments have been run, and the proposed benefits still need validation.